DOCKER FUNDAMENTALS

Q. Use Docker Compose to define and run multi-container applications?

Docker Compose is a powerful tool for defining and running multi-container Docker applications using a YAML file. With Docker Compose, you can specify the services (containers), networks, and volumes that make up your application, and then run them all with a single command.

Basic Steps to Use Docker Compose:

1. **Install Docker & Docker Compose**
   - **Docker Desktop includes Compose by default.**
   - **Confirm with:**

     bash

     docker-compose –version

2. **Create a docker-**compose.yml **File**
   - **This file defines the services, networks, and volumes for your app.**
3. **Run** docker-compose **Commands**
   - **Use** docker-compose up **to start everything.**
   - **Use** docker-compose down **to stop and remove the containers**

Example: Web App with Nginx + Flask + Redis

yaml

version: '3.9' # Compose file version

```yaml
services:
  web:
    build: ./web
    ports:
      - "5000:5000"
    depends_on:
      - redis

 redis:
   image: redis:alpine

nginx:
  image: nginx:alpine
  ports:
    - "80:80"
  volumes:
    - ./nginx.conf:/etc/nginx/nginx.conf:ro
  depends_on:
    - web
```

🔧 Directory Structure

markdown

```
project-root/
│
├── docker-compose.yml
```

├── nginx.conf

└── web/

├── Dockerfile

└── app.py

Common Commands

bash

 # Start all services

docker-compose up

 # Start in detached mode

docker-compose up -d

 # Stop all services

docker-compose down

 # Rebuild services

docker-compose up --build

 # View logs

docker-compose logs -f

Benefits of Docker Compose

- **Easy multi-container orchestration.**
- **Isolated development environments.**
- **Declarative configuration.**
- **Works well with CI/CD.**